PolypSegNet: An Attention-Guided Multi-Scale Feature Fusion Network for Accurate Colorectal Polyp Segmentation

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***Abstract*—Colorectal cancer (CRC) is a leading cause of cancer-related mortality worldwide, making early colorectal polyp detection essential for improving patient outcomes. However, accurate polyp segmentation remains challenging due to variations in polyp size, shape, and texture, along with low contrast, blurred boundaries, and imaging artifacts in endoscopic images. Although recent CNN-, Transformer-, and foundation-model-based approaches have improved segmentation accuracy, many remain limited by high computational cost and architectural complexity. To address these issues, this study proposes PolypSegNet, an efficient encoder–decoder architecture integrating a pretrained ResNet-50 encoder, Multi-Scale Feature Fusion (MSFF), and Convolutional Block Attention Modules (CBAM). The framework enhances multi-scale contextual learning and attention-guided feature refinement while maintaining computational efficiency. Experiments on the Kvasir-SEG dataset achieved a Dice score of 0.9164 and an IoU score of 0.8655, demonstrating competitive performance for colorectal polyp segmentation.**

***Index Terms*—Colorectal cancer, polyp segmentation, medical image segmentation, EfficientNet-B5, U-Net++, deep learning, colonoscopy, Kvasir-SEG.**

I. Introduction

Colorectal cancer (CRC) is one of the leading causes of cancer-related mortality worldwide, making early detection and removal of colorectal polyps essential for improving patient outcomes [1], [2]. Colonoscopy is the gold-standard screening method; however, accurate polyp segmentation remains challenging because polyps vary widely in size, shape, color, texture, and boundary appearance. Endoscopic images are also affected by low contrast, blurred boundaries, specular reflections, motion blur, and complex mucosal backgrounds [3], [4]. Although recent CNN-, Transformer-, and foundation-model-based approaches have improved segmentation accuracy, many advanced models require high computational resources, large parameter counts, or complex training pipelines, limiting their suitability for real-time clinical deployment [5], [6].

To address these limitations, this study proposes **PolypSegNet**, an efficient encoder–decoder framework for colorectal polyp segmentation. The model integrates a pretrained ResNet-50 encoder, a Multi-Scale Feature Fusion (MSFF) module, and Convolutional Block Attention Modules (CBAM) to enhance contextual representation and attention-guided feature refinement. The main contributions are:

- We propose an efficient ResNet-50-based encoder–decoder architecture for colorectal polyp segmentation.
- We introduce an MSFF module to aggregate multi-level encoder features and improve representation of polyps with diverse scales and appearances.
- We integrate CBAM-based channel and spatial attention in the decoder to refine discriminative features and improve boundary localization.
- The proposed model achieves a Dice score of 0.9164 and an IoU score of 0.8655 on the Kvasir-SEG dataset, demonstrating competitive segmentation performance with computational efficiency.

II. Related Work

A. Encoder–Decoder CNNs

U-Net and its variants, including ResUNet and ResUNet++, established the encoder–decoder paradigm for biomedical image segmentation [7], [8]. These models have been widely used for colorectal polyp segmentation and reported baseline Dice scores of approximately 0.788–0.810 on Kvasir-SEG.

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ResUNet++-Lite further improved this direction with masked Grad-CAM visualization, achieving 0.858 Dice and 0.791 IoU. However, conventional encoder–decoder CNNs often lose fine boundary details due to repeated pooling, while direct skip connections may transfer background noise from specular highlights, mucosal folds, and bowel content into the decoder.

B. Attention and Multi-Scale Methods

Attention mechanisms have been introduced to improve feature selectivity and suppress irrelevant background regions. PolypSEAG-Net integrates attention gates and squeeze-and-excitation blocks, achieving 0.8773 Dice and 0.8025 IoU on Kvasir-SEG [9]. DCATNet combines ResNetV2-50, a ViT bottleneck, deformable convolutions, and contextual attention, reporting 0.9351 Dice and 0.8845 IoU, but with higher computational demand [10]. Multi-scale feature fusion has also shown strong effectiveness for handling variable polyp sizes and appearances. PVT-MSFF-CNN combines a Pyramid Vision Transformer encoder with multi-scale feature fusion, achieving 0.910 Dice and 0.859 IoU [11], while DEGF-Net uses dual encoders with global-local feature aggregation and reports 0.933 Dice and 0.886 IoU [12]. These studies demonstrate the importance of attention-guided refinement and multi-scale contextual aggregation, although many require complex architectures or large GPU resources.

C. Foundation Models and Positioning of This Work

Recent foundation-model-based approaches, such as PSF-SAM, Polyp-DiFoM, and PolypDINO, have explored SAM and DINOv2 for polyp segmentation [13]–[17]. PSF-SAM achieves 0.9327 Dice and 0.8928 IoU but requires more than 100M parameters and over 400 GFLOPs, while Polyp-DiFoM reports Dice scores of approximately 0.912–0.917 through model distillation. Although these approaches show promising generalization, their computational requirements may limit real-time clinical deployment.

In contrast, the proposed PolypSegNet combines a ResNet-50 encoder, Multi-Scale Feature Fusion (MSFF), and CBAM-guided decoder to balance accuracy and efficiency. It achieves 0.9117 Dice and 0.8581 IoU with 25.94M parameters and 20.42 GFLOPs, surpassing comparable CNN baselines while remaining more computationally efficient than transformer-heavy and foundation-model-based methods.

III. Proposed Methodology

The proposed PolypSegNet is designed to improve colorectal polyp segmentation by combining hierarchical CNN feature extraction, multi-scale feature fusion, and attention-guided decoding.

As shown in Fig. 1, the proposed PolypSegNet follows an encoder–MSFF bottleneck–CBAM-attended decoder design. The ImageNet-pretrained ResNet-50 encoder extracts hierarchical feature maps, the MSFF bottleneck aggregates multi-level semantic information from deeper encoder stages, and the CBAM-attended decoder refines spatial and channel-wise features to generate the final polyp segmentation mask.

To improve multi-scale representation, feature maps from intermediate and deep encoder stages are passed to a Multi-Scale Feature Fusion (MSFF) module. Specifically, the selected encoder features are projected to a common channel dimension using 1×1 convolution, resized to the same spatial resolution, and fused through element-wise summation followed by a 3×3 convolution. This enables the network to capture both local boundary information and high-level semantic context.

The decoder progressively upsamples the fused feature representation to recover the segmentation map. Skip connections from earlier encoder stages are concatenated with decoder features to preserve spatial detail. Each decoder block applies convolutional refinement followed by a Convolutional Block Attention Module (CBAM), which enhances informative channel and spatial features while suppressing irrelevant background regions. Finally, a 1×1 convolution followed by a sigmoid activation generates the binary polyp probability map.

A. Convolutional Block Attention Module

CBAM is integrated into the decoder to improve feature refinement. It sequentially applies channel attention and spatial attention. In channel attention, global average-pooling and max-pooling descriptors are passed through a shared multilayer perceptron and combined to generate channel-wise weights. In spatial attention, average-pooled and max-pooled feature maps are concatenated and passed through a 7×7 convolution to produce spatial attention weights. This dual-attention mechanism helps the decoder emphasize polyp-relevant regions and reduce the effect of artifacts such as specular reflections and mucosal background structures.

B. Loss Function

The model is trained using a hybrid Dice and binary cross-entropy loss:

$$\mathcal{L} = 0.5\mathcal{L}_{Dice} + 0.5\mathcal{L}_{BCE}. \tag{1}$$

Dice loss improves region-level overlap and is effective for class-imbalanced segmentation, while binary cross-entropy provides stable pixel-wise optimization during training.

C. Training Protocol

All images are resized to 352×352 pixels and normalized using ImageNet statistics. The model is trained using the AdamW optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4}. A cosine annealing learning-rate schedule is used for 100 epochs with a minimum learning rate of 1×10^{-6}. Gradient clipping with a maximum norm of 1.0 is applied, and early stopping is performed with a patience of 15 epochs based on validation Dice score. Data augmentation includes horizontal and vertical flipping, 90° rotation, shift-scale-rotate transformation, elastic and optical distortion, color jitter, and Gaussian noise.

TABLE I
SUMMARY OF RECENT COLORECTAL POLYP SEGMENTATION STUDIES

Ref.	Author(s)	Dataset	Method	Dice	IoU	Other Metrics	Main Limitations
[18]	Yu et al.	KS, CL, CO, ES, ET	DEGF-Net: Swin-B + ResNet-18 dual encoder with global–local fusion	0.933 (KS), 0.958 (CL)	0.886 (KS), 0.922 (CL)	Prec.: 0.931/0.962; Rec.: 0.939/0.955	Weak on small, flat, and low-contrast polyps; no video or real-time validation.
[8]	Cristea et al.	C300, CL, CO, ET, KS, KSe, BKAI	PraNet with Res2Net backbone and augmentation study	0.511–0.926	0.387–0.872	MAE: 0.017; S: 0.964; E: 0.909	Single architecture; limited external testing and no cross-validation.
[19]	Mahbub et al.	KS, WCE-Neg	U-Net, SegNet, and ResNet18 with negative sample integration	∼0.95 val.	NR	Acc.: U-Net ∼0.95; SegNet ∼0.925–0.95	No explicit Dice/IoU table; no cross-dataset SOTA comparison.
[7]	Asare et al.	KS	PolypSeg-GradCAM: ResNet-34 U-Net with Grad-CAM	0.8902 ± 0.0125	0.8023 ± 0.0197	Sens.: 0.9058; Prec.: 0.9083; AUC: 0.9722	Single-dataset study; no video-based temporal validation.
[9]	Asare and Bagci	KS	U-Net with Soft Dice loss and Grad-CAM explainability	0.9612	0.9257	32.52M params; 50.902 GFLOPs	Weak on tiny/flat polyps; coarse boundary-level heatmaps.
[20]	Wu et al.	KS, CL, CO, PG	DepthPolyp with MiT-B0, pseudo-depth guidance, and uncertainty loss	0.853 (KS), 0.891 (N→C)	0.745 (KS), 0.805 (N→C)	Dice/IoU only	Depends on frozen depth estimator and uncertainty-weighted loss.
[21]	Liu et al.	CL, KS, PG	PolypSEAG-Net: lightweight CNN with AG and SE blocks	0.8855 (CL), 0.8773 (KS), 0.8103 (PG)	0.8204, 0.8025, 0.7694	Dice, IoU, Prec., Rec.	Training time grows on larger datasets; no transfer learning or MSFF.
[22]	Xiong and Sun	KS, CL, CO, ET, ES	DensePVT-Net with PVT encoder and boundary feature fusion	mDice: 0.924	mIoU: 0.873	mDice, mIoU, wDice, MAE, S-measure	Performance drops on ETIS and ColonDB; dense design adds complexity.
[10]	Agnihotri et al.	KS, CL, CO, ET, ES	Polyp-DiFoM: distillation from SAM, DINOv2, OneFormer, and Mask2Former	0.912–0.917 (KS)	0.861–0.864 (KS)	E-max: 0.956; MAE: 2.3 (KS)	Complex three-stage distillation; natural-to-endoscopy domain gap.
[11]	Rahman et al.	KS	ResUNet++-Lite with BCE + Dice loss and Masked Grad-CAM	0.858	0.791	Pixel Acc.: 0.964	Reduced robustness for tiny, flat, low-contrast polyps and artifacts.
[12]	Ali et al.	KS, PG-OOD	PolypDINO: frozen DINOv2 with LoRA and Mask2Former decoder	mDSC: 62.25–88.21	mIoU: 45.19–78.91	Rec.: 94.80%; Prec.: 93.25%	Large cross-center variation; decoder-dependent performance.
[13]	Wang et al.	KS	ConvNeXt encoder with transformer decoder and cross-attention skips	0.8715 test, 0.8802 val.	0.8021 test, 0.8120 val.	Prec.: 0.9023; Rec.: 0.8796	Single-source evaluation; high latency limits real-time use.
[15]	Kerkaou et al.	CL, KS; CO, ET, C300	PVT encoder with MFM, CRA, and CNN decoder	0.939 (CL), 0.910 (KS)	0.890 (CL), 0.859 (KS)	Dice/IoU only	Low-contrast flat or sessile polyps remain challenging.
[23]	ELKarazle et al.	KS, CL; ET, CO, C300, BKAI	PolyNeXt: ConvNeXtV2 encoder with FCN decoder	0.980–0.991	0.962–0.982	Acc.: 0.980–0.991	Weak near colonoscope lens and specular reflections; clinical FPS untested.
[24]	Qayoom et al.	KS, CL, CO, ET, ES, C300, BKAI	Survey of CNN, transformer, hybrid, and ensemble methods	0.79–0.955	0.782–0.915	DBMF Acc.: 0.973; Prec.: 0.959	Small/flat polyps remain difficult; skip connections may pass noise.
[17]	Wang et al.	KS, CL; CO, ET, C300	DCATNet: ResNetV2-50, ViT, deformable conv., and attention modules	0.9351 (KS), 0.9444 (CL), 0.7963–0.9015 OOD	0.8845 (KS), 0.8980 (CL), 0.7022–0.8354 OOD	Prec.: 0.9423; Rec.: 0.9388 (KS)	Performance drops on tiny/flat polyps; complex hybrid design.
[25]	Wang et al.	KS, CL	PSF-SAM: frozen SAM with multi-scale polyp feature tokens	0.9327 (KS), 0.9354 (CL)	0.8928 (KS), 0.8943 (CL)	Prec.: 0.9412; Rec.: 0.9304	Sensitive to token initialization; gains reduce with larger training sets.
[26]	Ouria et al.	KS, CL, ET	SegFormer-B4 with adaptive fusion and edge-aware loss	0.946 (KS)	0.899 (KS)	Dice/IoU only	Poor low-contrast performance; inpainting adds latency; no video consistency.

Note: KS = Kvasir-SEG; CL = CVC-ClinicDB; CO = CVC-ColonDB; ET = ETIS/ETIS-LaribPolypDB; ES = EndoScene; C300 = CVC-300; KSe = Kvasir-Sessile; BKAI = BKAI-IGH; PG = PolypGen; PG-OOD = PolypGen out-of-distribution test; WCE-Neg = WCE curated negative colon images; AG = attention gate; SE = squeeze-and-excitation; MFM/MSFF = multi-scale feature fusion; CRA = convolution refinement attention; NR = not reported.

IV. EXPERIMENTAL SETUP

A. Dataset

Experiments are conducted on the Kvasir-SEG dataset, which contains 1000 colonoscopy images with corresponding pixel-level binary polyp masks. The images vary in resolution and include polyps with diverse size, shape, and appearance. The dataset is divided into 70% training, 15% validation, and 15% testing sets, corresponding to 700, 150, and 150 images, respectively. A fixed random seed of 42 is used to ensure reproducibility.

B. Evaluation Metrics

Model performance is evaluated using Dice coefficient, Intersection over Union (IoU), precision, recall, F2-score, and mean absolute error (MAE). All metrics are computed per image at a threshold of 0.5 and averaged over the test set. Dice and IoU measure segmentation overlap, precision quantifies false-positive control, recall measures missed-polyp reduction, and F2-score emphasizes recall, which is clinically important for minimizing missed lesions.

C. Baseline Models

To ensure a fair comparison, the proposed model is compared with plain encoder–decoder baselines using seven ImageNet-pretrained backbones: ResNet34, ResNet101, VGG16, EfficientNet-B4, MobileNetV3-Large, DenseNet121, and InceptionV3. These baselines do not include the MSFF module or CBAM attention. All models are trained using the same preprocessing, augmentation, optimizer, loss function, and training schedule.

D. Sensitivity Analysis

A sensitivity analysis is performed to examine the effect of input resolution and random initialization. Additional experi-

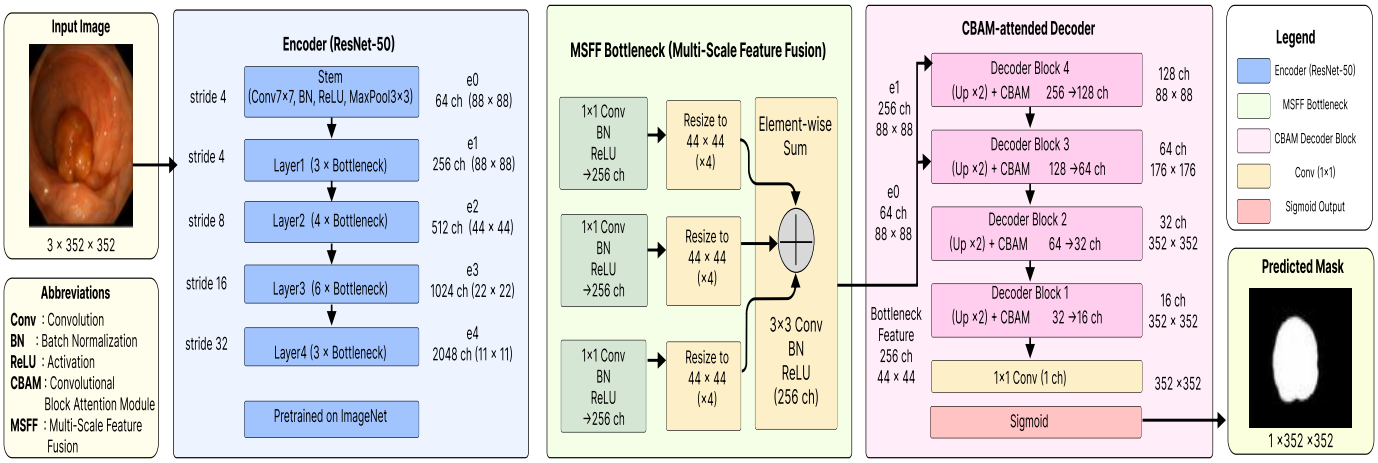


Fig. 1. Proposed PolypSegNet architecture. The framework consists of an ImageNet-pretrained ResNet-50 encoder, a Multi-Scale Feature Fusion (MSFF) bottleneck that fuses encoder features from e_2, e_3, and e_4, and a CBAM-attended decoder with skip connections from e_1 and e_0 for accurate colorectal polyp segmentation.

TABLE II
BASELINE COMPARISON ON KVASIR-SEG.

Model	Dice	IoU	Prec.	Recall	F2	MAE
ResNet34	0.8961	0.8308	0.9102	0.9127	0.9017	0.0360
ResNet101	0.8936	0.8284	0.9063	0.9137	0.9015	0.0400
VGG16	0.8852	0.8145	0.9008	0.9024	0.8893	0.0418
MobileNetV3	0.9039	0.8437	0.9080	0.9222	0.9111	0.0347
EfficientNet-B4	0.9065	0.8452	0.9069	0.9292	0.9156	0.0336
Proposed	0.9117	0.8581	0.9134	0.9376	0.9205	0.1061
Proposed + TTA	**0.9164**	**0.8655**	**0.9142**	**0.9464**	**0.9279**	0.1066

TABLE III
COMPARISON WITH RECENT METHODS ON KVASIR-SEG.

Method	Dice	IoU
ResUNet	0.788	0.778
ResUNet++	0.810	0.790
ResUNet++-Lite+XAI	0.858	0.791
PolypSeg-GradCAM	0.890	0.802
PolypSEAG-Net	0.877	0.803
HarDNet-MSEG	0.904	–
PVT-MSFF-CNN	0.910	0.859
Proposed	0.9117	0.8581
Proposed + TTA	**0.9164**	**0.8655**
EENet	0.921	0.866
PSF-SAM	0.933	0.893
DEGF-Net	0.933	0.886
DCATNet	0.935	0.885
SegFormer	0.946	0.899

ments are conducted using input sizes of 128×128, 224×224, and 256×256 pixels with random seeds 37, 42, and 52. These settings are compared with the main configuration of 352×352 pixels and seed 42.

E. Implementation Details

The proposed model is implemented in PyTorch and trained on an NVIDIA Tesla T4 GPU. FLOPs are calculated using the `thop` library at an input size of 352×352. Inference time is measured with batch size 1 as the average of 50 runs after 10 warm-up iterations using CUDA synchronization.

V. RESULTS AND DISCUSSION

A. Baseline Comparison

Table II compares the proposed model with standard encoder–decoder baselines. PolypSegNet achieves the best Dice (0.9117) and IoU (0.8581), outperforming the strongest baseline, EfficientNet-B4, by 0.52% Dice and 1.29% IoU. Applying test-time augmentation (TTA) further improves performance to 0.9164 Dice and 0.8655 IoU without additional training cost, demonstrating the effectiveness of the MSFF and CBAM modules.

B. Comparison with State-of-the-Art Methods

Table III compares PolypSegNet with recent polyp segmentation approaches. The proposed model surpasses conventional CNN-based methods and achieves performance comparable to transformer-based and foundation-model approaches while requiring substantially fewer computational resources.

TABLE IV
ABLATION STUDY.

Variant	Dice	IoU	Acc.	F2
Baseline	0.8962	0.8350	0.8808	0.9056
No MSFF	0.8996	0.8360	0.9682	0.9087
No CBAM	0.8937	0.8266	0.9677	0.8886
No Pretrained	0.6271	0.5124	0.9007	0.6564
Full Model	**0.9039**	**0.8437**	**0.9695**	**0.9111**

C. Ablation Study

Table IV evaluates the contribution of each component. Removing ImageNet pretraining causes the largest performance degradation, confirming the importance of transfer learning for limited medical datasets. Both MSFF and CBAM contribute to improved segmentation accuracy, with the complete model achieving the best overall performance.

TABLE V
SENSITIVITY ANALYSIS.

Seed	Size	Dice	IoU	F2	MAE
37	128	0.8732	0.8035	0.8840	0.1333
42	224	0.9102	0.8558	0.9234	0.1219
52	256	0.8844	0.8275	0.8927	0.0786
42	352	**0.9117**	**0.8581**	**0.9205**	0.1061

TABLE VI
MODEL COMPLEXITY AND TRAINING CONFIGURATION.

Metric	Value
Parameters	25.94 M
FLOPs	20.42 GFLOPs
Inference Time	25.22 ms
Throughput	39.65 FPS
Backbone	ResNet-50
Input Size	352×352
Loss	Dice + BCE
Optimizer	AdamW
Learning Rate	1×10^{-4}
Early Stopping	15 epochs

D. Sensitivity Analysis

Table V summarizes the effect of input resolution and random initialization. Higher image resolution consistently improves segmentation performance, while seed variation introduces only minor fluctuations, indicating robust model behavior.

E. Training Dynamics

Figure 2 illustrates the training process. Training and validation losses decrease smoothly, while Dice and IoU steadily improve and converge without significant overfitting, indicating stable optimization and effective regularization.

F. Qualitative Results

Figure 3 presents representative segmentation examples. The proposed model accurately localizes polyps of varying sizes and morphologies. Grad-CAM visualizations indicate that the network focuses on clinically relevant polyp regions, while error maps show that residual errors are mainly concentrated around object boundaries.

G. Model Complexity

Table VI summarizes the computational profile of PolypSegNet. The model achieves competitive segmentation accuracy while maintaining low computational cost, making it suitable for real-time deployment.

VI. LIMITATIONS AND FUTURE WORK

Despite promising results, this study has several limitations. First, experiments were conducted only on the Kvasir-SEG dataset; therefore, cross-dataset generalization to datasets such as CVC-ClinicDB, CVC-ColonDB, ETIS-LaribPolypDB, and EndoScene remains to be investigated. Second, residual segmentation errors are primarily observed near polyp boundaries, particularly for large and irregular lesions. Third, the proposed framework was evaluated on static images and has not been validated on colonoscopy video sequences or real-time clinical workflows. Future work will focus on cross-domain evaluation, integration of boundary-aware learning strategies, video-based segmentation, and statistical significance analysis across multiple experimental runs.

VII. CONCLUSION

This paper presented PolypSegNet, an efficient colorectal polyp segmentation framework that combines an ImageNet-pretrained ResNet-50 encoder, Multi-Scale Feature Fusion (MSFF), and Convolutional Block Attention Modules (CBAM). The proposed architecture effectively captures multi-scale contextual information while refining discriminative spatial and channel features. Experimental results on the Kvasir-SEG dataset achieved Dice and IoU scores of 0.9117 and 0.8581, respectively, which further improved to 0.9164 and 0.8655 with test-time augmentation. Comparative experiments demonstrated that PolypSegNet outperforms several conventional CNN-based baselines while maintaining substantially lower computational complexity than many transformer-based approaches. Ablation studies confirmed the complementary contributions of MSFF and CBAM, while sensitivity analysis demonstrated stable performance across different experimental settings. These findings indicate that lightweight attention-guided multi-scale feature learning provides an effective and computationally efficient solution for colorectal polyp segmentation and shows potential for future computer-aided diagnostic applications.

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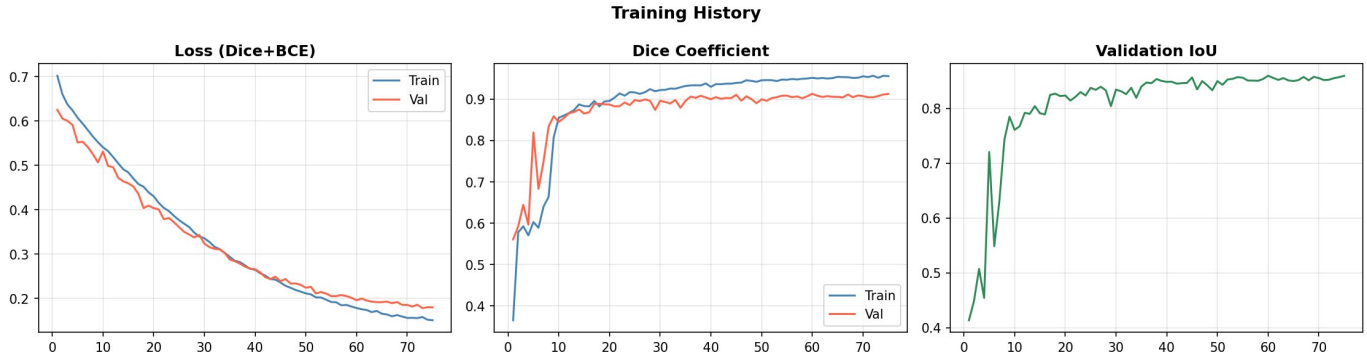


Fig. 2. Training loss, validation Dice, and validation IoU across epochs.

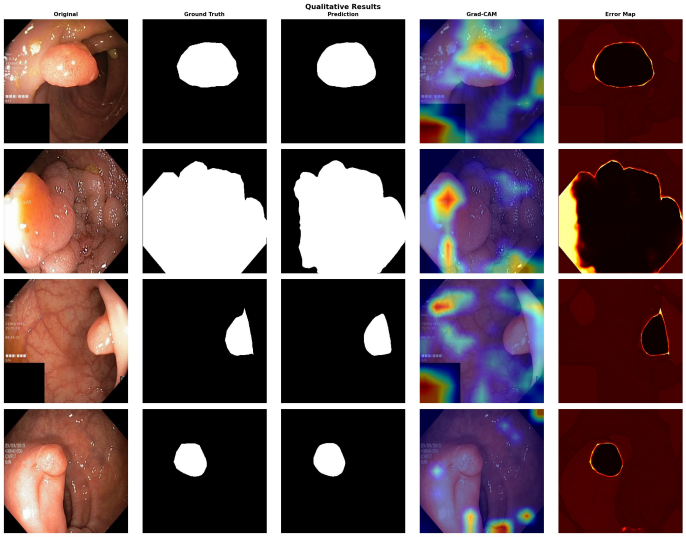


Fig. 3. Qualitative results showing original image, ground truth, prediction, Grad-CAM, and error map.

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